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Ethical concerns for theoretical research in linguistics: issues and best practices

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Abstract: Although formal, e.g., generative, approaches to linguistic inquiry are primarily concerned with the analysis of structural patterns, ethical concerns for the elicitation and documentation of empirical data that motivate these analyses persist. In this article, we explicate the challenges to be met and responsibilities that formal linguists must take in data collection, documentation, and analysis to ensure high ethical standards. We direct our attention especially toward research on heritage, minoritized and endangered languages, which have enriched the theoretical landscape in recent decades. Finally, we show that, when proper ethical concerns are recognized and accounted for in formal research, this collective body of research in its current form is also fully capable of respecting the integrity of individual speakers and their communities.

Keywords: ethics; generative grammar; heritage Germanic; heritage Italian dialects; Rajbanshi; heritage linguistics

1 Introduction

This paper deals with ethical issues in theoretical linguistics, intended as a branch of linguistics whose primary aim is to build a theory of language and a formalization of the laws underlying grammars. The considerations that we draw are however restricted to the generative sub-branch of theoretical linguistics, a.k.a. Generative Grammar (GG) or Generative Linguistics (GL henceforth), initiated by Chomsky (1953). Other theory-oriented approaches, such as LFG (Lexical Functional Grammar [Kaplan and Bresnan 1982]) or Dependency Grammar (Tesnière 1959), will not be considered simply because the expertise of the authors lies in proof-theoretical, derivational approaches (as compared with model-theoretical, representational

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ones). Furthermore, the article will mainly consider issues related to the study of syntax, and therefore all approaches addressing the study of phonology, semantics, or pragmatics from a formal-theoretical viewpoint will not be taken into account here. The goal is to offer a viewpoint on ethical issues related to a branch of linguistics that mainly focuses on formalizing the laws of language, by documenting our experience in the field. Obviously, the paper is not exhaustive; it only highlights some issues that we believe are important. Given that many of the data collection techniques are the same as those used by other approaches, a majority of these issues will not be restricted solely to GL. We will try to highlight the ones that are.

Traditionally, theoretical linguistics is a branch of research with a focus on structural patterns underlying mental grammars/(individual) languages of speakers. It is often assumed that because of the nature of its inquiry, generative linguists concentrate solely on the grammar of individual speakers, neglecting the geo-political, historical, and cultural aspects shaping languages and linguistic communities. While language communities are not the primary object of the study of GL, and indeed for the initial decades since its inception in the 1950s, intuitions from a single speaker were considered enough to garner insights into the workings of Universal Grammar (UG), there is now a growing interest among generative researchers to work with larger data samples including those from previously under-explored languages.

An increased focus on linguistic diversity translates into an increase of research on unexplored languages, including dialects, minoritized and heritage varieties. This shift brings about the need to reflect on the standards and ethical methods for data collection, description, and the distribution of findings stemming from this research. In summary, the more GL researchers distance themselves from English and other standard languages, the more attention they must pay to under-represented languages, resulting in the need for more stringent attention to data collection methods. While the attention towards under-explored languages and variation has increased, ethically sound methodology for data collection and analysis has received almost no attention in the field. Indeed, many of the most important reference works on ethics in linguistics at large are concerned with applied linguistics or sociolinguistic practices (BAAL 2021; Silver et al. 2023), while almost nothing is said about more theory-oriented approaches (but see Himmelfmann 1998).

Our article highlights and critically evaluates the main issues and best practices in data collection in current GL-based research, with a focus on heritage and minority varieties of Indo-Aryan, Romance, and Germanic. We first list a number of methodological points that are particularly good, to our view, in the GL methodology. Then, we move on to consider more problematic issues. In line with what has already been claimed by other authors, most notably Bender (2019) and D'Arcy and Bender (2023) for Natural Language Processing (NLP), we notice some serious shortcomings

having to do, for instance, with name attribution, or acknowledgment (or lack thereof) of the speaker's perception of their language. Most theoretical work targets grammars and grammatical systems, and therefore not language as a whole. While we concede that grammar, intended primarily as an abstract set of rules, can be considered independent of its users (see Czaykowska-Higgins 2018; Dobrin and Berson 2011; Dobrin and Schwartz 2021), we take issue with the general assumption according to which all sociolinguistic related information, as well as the attitude of the speakers, can be completely dismissed by GL. We show that an ethically grounded methodology is also favorable to better science; therefore ethics is not only a moral obligation for researchers, but it also gives a scientific added value to research.

We acknowledge and expect variation at the individual- and community-level with respect to structures produced and those evaluated to be 'acceptable'. We mention this since a point of clarification regarding our research program concerns the role of sociolinguistic factors, such as register, in GL research. Although vital in capturing a holistic understanding of language competence and performance, these factors will almost certainly, not (radically) affect argument structure and event semantic requirements. Therefore, we will not be overly concerned with issues of register, given that the structures of interest to syntax cannot be modified or influenced externally. Their choice certainly can be, and we indeed consider every structure as a possible output of human grammar, but the mechanisms generating a sentence cannot be.

1.1 Ethics and GL: major issues and main solutions

This paper focuses on some ethical issues that concern the methodology of GL. Given that GL is obviously an approach on languages, many of the concerns that GL practitioners experience are shared with the large community of linguists. Some of the concerns of GL practitioners can be experienced equally by community-engaged data collectors. In general, the authors are not claiming here that what they say only holds for theory-oriented linguistics: they simply engage with their field of research, leaving the application of the same reflections for every other field where these might be relevant. Furthermore, since the authors are GL practitioners, they are obviously more knowledgeable about this specific framework, but the observations and concerns regard the theoretical study of linguistics in general. An additional disclaimer is in order here: GL is a theoretical framework in that it aims to discover the laws underlying language and formalize them. Claiming that GL is theoretical does not imply in any way that other subfields do not have a theory or a formalism; GL has formalization and the creation of a theory as its main aim (as opposed to, for

instance, language description), and therefore the term “theoretical” describes it well.

The rest of the article is structured as follows: Section 2 will deal with ethical issues that we believe to be tackled rather positively in GL, offering examples from various sources, but mainly from the authors’ experience. Section 3 will deal instead with some shortcomings and drawbacks of the GL approach, in regard to the (often) complete absence of relevant sociological information about linguistic communities, that directly or indirectly affects data collection and linguistic analyses. Section 4 will present our conclusions.

2 The ethics of GL: some positive sides

Since its inception, research carried out within the generative tradition was centered on Noam Chomsky’s initial attempts to formalize structural aspects of natural language (Chomsky 1957, 1965). Elements of these initial works are still present in computational approaches, where sequences of rules apply to structural elements (syntax) and are further implemented in formal computational notation (NLP [Berwick and Stabler 2019]).

This formal approach to linguistic analysis, known as *transformational grammar*,¹ advanced the position of a universal underlying structure, i.e., the syntax, of natural languages. The advantage of this architectural assumption is that it reduces natural languages to a set of units (here, strings) obtained by combining elements from a lexicon through the application of rules that manipulate these units. Approaching linguistic research from this perspective enabled clearer definitions of fundamental concepts. For example, *languages* refer to a set of strings in some finite set of symbols (called the *vocabulary* of a given language), while a *grammar* classifies a set of rules that provides a recursive enumeration of the strings belonging to the language (Chomsky and Schützenberger 1963: 118). This narrow focus on the formal nature of strings unfortunately did not acknowledge or include variation within and among natural languages.

As research in this tradition continued, its adherents recognized that the extant forms of language are a subset of theoretically possible structures, though a

¹ We recognize that there is a separate family of *generative* approaches that are declarative (rather than transformational) in nature; see, e.g., Borsley and Börjars (2011) for an overview of the core desiderata of these frameworks. We do not discuss these frameworks further in this chapter; however, the concerns surrounding large-scale data collection, in particular, and ethical research in formal linguistics, in general, also hold for these non-transformational ones.

theoretical way to capture this subset was not immediately evident. Government and Binding (GB) Theory (Chomsky 1981) provided a workable solution to this central problem by postulating a set of universal principles of grammar construction (a.k.a. UG), each of which had an open parameter, the value of which was determined, or *set*, based on an individual's experience and exposure to principal linguistic data; i.e., the Principles and Parameters (P&P) architecture. Because of its accuracy, GB opened the path to research in languages beyond a select few, especially thought distinct from English (Fukui 1986; Huang 1982). Since the early 1990s, many generative linguists have continued to carry out their formal research based on architectural assumptions in the Minimalist Program (MP) (Chomsky 1995). Although many features and axioms from the GB era have been reconfigured in the MP, the additional aim of the latter is to approach language from the perspective of biological endowment. This purportedly assists us in better understanding the evolutionary reason for language design and the relation that our human language faculty (narrow) may have evolved from primate cognition and communication systems (broad). Continued research from a Minimalist perspective has a broad empirical and comparative focus, which has led to a realization of the importance of developing formal approaches to grammar that both acknowledge, describe, and potentially predict the presence of linguistic variation (cf. Kayne 2000; Ledgeway 2000; Polinsky 2018).

2.1 Foundational elements of GL methodology

In spite of the shift in focus from developing a purely combinatorial formal system to one that is driven by questions of optimal derivation and design (cf. Boeckx and Hornstein 2010), the fundamental goals of generative research remain the same.

The primary objectives of GL are to build an acquisition model and reveal abstract linguistic representations underlying the linguistic competence of native speakers (Becker and Ud Deen 2020). To this aim, the *ideal speaker–hearer* is a central (and often misunderstood) concept in generative research. The ideal speaker–hearer refers to an individual who is abstracted from the surface variations and oscillations that characterize a speech community. Linguists rely on the intuitions and meta-cognitive judgments of native speakers that are reflections of their underlying linguistic knowledge. These concepts demand some re-evaluation, once we acknowledge language varieties, which, allegedly, differ in structure and form from conventionally considered standard languages of monolinguals, and whose speakers may or may not display the same level of linguistic competence as monolinguals. In the next section we will briefly discuss the ethical implications of referring to “idealized speakers” for linguistics research.

2.2 The idealized speaker

The speaker is considered by GL mainly as a sort of language machine operating a grammar on a vocabulary. With the spotlight remaining solely on the abstract grammar of individual speakers, the researcher is able to sidestep the complex identity issues defining both the speakers and their languages. It is a recognized fact that individuals have their social and community affiliations; neither they nor the languages they speak are disembodied, abstract constructs (see, for example, Dobrin and Berson 2011; Czakowska-Higgins 2018: 110–121). However, when the objective is to examine the mental representations underlying linguistic knowledge of speakers, these complex socio-cultural factors act to impede rather than facilitate such inquiries. While the conception of idealized speakers, speech communities and linguistic forms have their own challenges, they help in maintaining scientific objectivity, to a certain extent, in linguistic research. In this sense, GL is neither focused on the speaker's feelings about their own language, nor on their self-perception, nor on the societal aspect of language.

While these factors do not have a straightforwardly identifiable impact on I-language (intended as the internal representation of a speaker's grammar), recognizing that the I-language of one speaker is different from that of another is an important methodological point. Whether this entails anything for the ethical stance of this methodology, however, remains to be seen. A related question is how ethical is it to research based on abstractions, when we are in the presence of human beings? One easy answer would be that every science needs to abstract away from the here and now and idealize a situation in order to uncover the underlying laws that govern it. However, it is one thing to create an idealized body to investigate Galileo's law, but quite another to create an idealized human being when it comes to investigating their internal grammar. In fact, the concept itself of a monolingual speaker is flawed in many ways and often carries a connotation of superiority that is highly problematic. A recent paper by Leivada et al. (2023) outlines the various misconceptions associated with the use of the concept of monolingual speakers in linguistic research; while the article is mainly targeted at psycholinguistic practice, where a monolingual control group is often considered essential for conducting any study, the general idea that monolinguals possess a "pure" language against which all other varieties must be checked is commonly shared across other branches of linguistics, including GL, to some extent.

One important point to be made in this respect is that, despite their best intentions, researchers also have their own biases, cultural viewpoints, and value

judgments.² In order to carry out a proper analysis of the interaction between culture and grammar, much more information and background knowledge is necessary than what can be found in most typical studies in the generative tradition. Abstracting away from the social milieu in which the data are used by speakers is useful because it creates the right conditions where the researcher is trained and expected to treat the data in the same unbiased manner as they treat their own linguistic judgments.

In fact, one of the primary goals of generative linguistics since its inception has been to describe intuitions, not to prescribe rules to its speakers; see for instance, Chomsky's (1954) criticism of the correctional approach adopted by Reiger (1953) in his textbook of Modern Hebrew. Language description, albeit within the confines of a theoretical framework, is a norm in generative studies, and it is very rare that generalizations and theoretical predictions are formulated based on speakers' socio-cultural status or racial and gender identities.

This statement, controversial as it may seem, rests at the core of GL enquiry, where ensuring an equal representation of all members of a community is not really the main concern. This is, however, not because GL practitioners do not care about the speakers or minorities: rather, it is because while many aspects of language can be influenced and impacted by external intervention, syntactic structure rarely is. One notable exception to this generalization is the dialect of Pescasseroli, described in a number of articles by Mario Saltarelli (Neri and Saltarelli 2005; Saltarelli 1999). In Pescasseroli, two determiner and demonstrative systems coexist, and they are linked to the gender of the speakers who use them: the men use one system, the women in the community use a different one. In cases like this, what GL does is consider the two separate systems as possible language systems, with no further need for specification. This last comment opens the path to another issue very often contested with regard to GL, namely acceptability judgments, and who decides what is acceptable. We will return to this in the next section.

The objective stance of GL is maintained even when analyzing large data-sets such as corpora. Corpora are treated and analyzed as a collection of linguistic outputs of a larger group of speakers; the objective is still to reveal the internal mechanisms of the language machine. Individual attitudes and perceptions, and the collective beliefs and practices of the community are not taken into account in GL analysis of corpora (even if they are contained there, like in the case of the New Mexico Spanish–English bilingual corpus at Penn State University also described in

2 One very central aspect of acknowledging this is recognizing the colonialist bias in data collection (see D'Arcey and Bender 2023; Gal and Irvine 1995).

Cacoullos and Travis 2018),³ and they do not influence the scientific analyses of the linguistic data.

2.3 Standardized and non-standardized languages

Aiming at scientific objectivity is in principle a good way to proceed for the sort of study that GL wants to carry out. However, it crucially relies on an idealized speaker, as stated above, and on the implicit idea that a community of speakers all share a common grammar. While this is mostly true for so-called macro-parameters, such as word order, null subjecthood, and morphological typology, the concept of uniformity of the speaker breaks down as soon as we pay attention to variation. There are two kinds of variation that might not depend on strictly grammatical factors but do have an impact on grammar: the variation concerning registers, and the variation concerning standardization. Registers are very rarely considered in GL studies, which are mainly carried out with the underlying idea that every register is a different language. The second form of variation, related to the first one, concerns standardization: non-standardized varieties show less uniformity and more optionality, and this kind of internal variation is rarely taken into account by GL.

One typical way to tackle this kind of variation is to consider quantitative analyses, based on the collection of a sufficient number of examples so as to make it possible to draw statistics-based generalizations. If a phenomenon is significantly present in a language, it is worth investigating. If not, it is possibly an exception, an accident, and we are not allowed to draw any generalizations based on one or two examples taken from one or two speakers. This approach has the advantage of being, at least superficially, more objective, but at the same time runs the risk of creating a research targeting only major and standardized languages. We will return to this in Section 2.6.

One objection that many raise regarding what has just been discussed concerns the fact that what might look like genuine intuitions is in fact a rule prescribed by some socially dominant group. That a speaker shows that they have an intuition or a judgment on a grammatical expression does not mean that they have not been influenced or pushed into that intuition. This state of things often characterizes community-based data collection, especially when languages are targeted where a more educated speaker is present along with other informants. In some rural communities in India, for instance, fieldwork conducted by one of the authors consisted of a group of informants gathering in a common space for an interview

³ *New Mexico Spanish–English codeswitching database*, Penn State University, <https://nmcode-switching.la.psu.edu/>. We thank a reviewer for pointing out this resource to us.

with the fieldworker. Sometimes, individual variants would get sidelined in the din of group conversations and more authoritative voices (see Gumperz 1958). On one such trip, the younger and educated speakers of Bangla (Eastern Indo-Aryan) dialects, for example, were seen increasingly borrowing honorific pronouns from standard Bangla. These are typically absent in the Bangla dialectal space, where close-knit communities use informal, non-honorific terms for all (irrespective of status, age, and gender). In the presence of educated community members, the dialectal intuitions may be suppressed as unpolished variants by the speakers themselves. This is not only ethically dubious but also creates a problem for theory-building, where extant grammatical possibilities escape investigation because the speakers are not confident enough to present them to the fieldworker.

More trivially, forcing the speaker's own intuitions in a given direction also includes the cases of standardized languages, where an external authority, be it an academy or just literary tradition, has decided which form is acceptable and which is not in a grammar (see Bauer 1998; Wolfram 1998, and many others). It is often the case that speakers unconsciously reproduce the standard acceptable patterns (Gordon 1997), which results not only in flawed data collection, but also in an unethical enforcement of a version of the language decided by someone who is not the speaker.

2.4 The speaker's intuitions

GL aims at devising a set of procedures that enable linguists to deduce statements about a 'linguistic grammar' from a fixed sample of linguistic material from any given language. It does so by trying to identify the 'limits' of grammar, by investigating borderline constructions, the acceptability of which can show whether a given grammar can or cannot generate them. Investigating the limits of variation entails checking for the acceptability of sentences that would not be normally produced spontaneously, because of their complexity. Metacognitive evaluations of elements of grammar also help linguists achieve this goal.⁴

There are a number of potential ethical concerns associated with this methodology, among which are the ones we stated in the previous section; however, one clear advantage is that it makes the individual speaker and their I-language the central unbiased focus of the research. Generative methodology puts the speaker's intuitions at its core: the speakers are, for GL, the ultimate authorities and the

⁴ Intuitions about language are considered in the formal, Aristotelian sense, as abstract rules that apply to some matter, independently of the matter itself (e.g., rules applying to a set of words, like adjectives, independently of the adjectives themselves).

owners of their data, which is ethically speaking a very inclusive strategy. Whether the speaker is also the legal owner of the data is an important issue. Legally, like in every other research involving humans, the researcher needs to observe data protection protocols, granting the owner permission to retract their consensus to data usage at any time for instance. Different countries have different regulations however, and some do not have them at all. This creates some disparity and restricts the possibility to collaborate across countries: many data protection protocols include a requirement to collect the data, also in third countries, under the supervision of a data protection officer. However, some institutions do not have such a figure, which creates major problems in carrying out international cooperations, very often to the detriment of researchers located in poorer countries. While GL particularly relies on this kind of elicitation, collecting data in different countries using different protocols is a common challenge for any linguist involved in community-engaged data collection. Therefore, this is a broader issue that concerns the entire linguistic community.

The speaker's intuition is of equal importance to the frequency in forms found in language production. An illustration of this point is Hopp and Putnam's (2015) study of the placement of finite verbs in subordinate clauses in Moundridge Schweitzer German (MSG). This moribund variety of German is spoken in south-central Kansas by a remaining two dozen speakers (with significant proficiency). In their study, Hopp and Putnam (2015) sought to test whether or not the asymmetric V2-pattern, i.e., with the finite verb in second position in declarative matrix clauses and as the final element in subordinate clauses, was still present in MSG. The expectation was that V2 would be weakened in main clauses, as reported in other studies on similar varieties such as heritage Danish (Kühl and Petersen 2018) and heritage Norwegian (Westergaard et al. 2023), and that the embedded clauses would keep the verb-final configuration. Surprisingly, however, Hopp and Putnam (2015) observed a split in the preference of finite verb placement in subordinate clauses in MSG. Based on raw production data from the eight speakers who participated in this study, subordinate clauses headed by *dass* 'that' (1a) and *weil* 'because' (1b) appear consistently with the V2-order (as found in matrix clauses), while those that begin with temporal adverbial clauses (1c) and relative clauses (1d) existed in the V-last pattern:

- (1) a. ... *dass da lieber Gott **hot** uns auch net alles **genomm** wie*
 that the dear God has us also not everything taken like
 dat in Oklahoma.
 there in Oklahoma
 '...that (the) dear God hasn't taken everything away from us like in
 Oklahoma.'
- b. ... *weil ich **duh** nett Hochdeutsch **redde**.*
 because I do/can not High German talk
 '... because I can't speak standard German.'

Table 1: Mean acceptability judgments and standard deviations.

Condition	Mean rating	Standard deviation
SOVAux	2.46	0.74
OVSAux	2.54	0.80
*N _{NOM} N _{NOM} VAux	2.58	0.99
*N _{NOM} N _{ACC} VAux	2.54	0.82
*SAuxAdvOV	5.08	0.77
*OAuxAdvSV	4.85	0.92
*AdvAuxSOV	5.19	0.64
*SAuxVOAdv	2.94	1.46
*Word salad	1.69	1.09

- c. ... wenn mir erscht **geheirat henn.**
when we first married have
‘... when we first got married.’
- d. ... die wo in die Schul jetzt **sin.**
those RP in the school now are
‘... those who/that are in school currently.’

At this point, an acceptability judgment contrasted with what the speakers were expected to do, as heritage language speakers, namely weaken V2 in matrix clauses, certainly not enforce V2 in embedded ones. The way in which Hopp and Putnam (2015) proceeded was to take the intuitions and the production of the speakers very seriously rather than dismissing them as not in conformity with the expectations. Therefore, Hopp and Putnam (2015) proceeded to design an acceptability judgment task to test whether the remaining speakers of MSG would evaluate a subset of these subordinate clauses with V2-ordering as acceptable.

The results of this acceptability judgment task are reported in Table 1 (Hopp and Putnam 2015: 201).

Examples (1a) and (1b) are worth emphasizing, because a group of minority speakers were presenting a “deviant” grammatical behavior with respect to the rest of Heritage Language (HL) speakers. Accepting these results at face value, Hopp and Putnam (2015) claim that the remaining speakers of this extra-territorial variety of German have created a more complex grammar with respect to the placement of finite verbs in subordinate clauses.

2.5 One speaker is enough

Data in GL are collected from speakers through observation, though in most cases linguists elicit data through questionnaires and one-on-one interviews. There are

also no specific guidelines for the number of consultants required for any study. It is understood that an entire corpus cannot capture the entirety of all possible sentences generated by a single speaker or a group of speakers, since they are, in principle, infinite in number. This particular issue is called *the data amount challenge* by D'Alessandro et al. (2021), because it calls into question just how much data is required to conclusively demonstrate that a particular linguistic phenomenon is 'part of the grammar' and worthy of detailed analysis. Given the relatively small number of remaining speakers with high, and even mediocre, levels of proficiency, it is often difficult to find a number of informants that achieves a measure of 'statistical significance'. Three examples confirm this point: pioneering research on the phonology of Frenchville French (spoken in Pennsylvania) by Bullock and Gerfen (2004a, 2004b, 2005) bases its claims on data from two speakers, and as we saw in the previous section, Hopp and Putnam (2015) were only able to recruit eight informants who were able to participate in both the production and judgment tasks in their study on finite verb placement in MSG. Furthermore, the ERC project *Microcontact*, targeting heritage Italo-Romance dialects in the Americas, had only a handful speakers per variety.

The basic GL methodology is focused on answering the question *What is a possible structure and what isn't?* This means that if a structure is systematically produced by even a single speaker, it is considered a possible output of natural language, and is, therefore, worth investigating. Replicability can obviously be a concern in this respect, which is why the standard methodology involves consulting more than one speaker. However, in principle, if one speaker consistently produces a structure, that structure is relevant for GL inquiry. With its focus on what the well-formedness of structures tells us about the overall grammar, these structures remain the object of scientific inquiry. In this respect, GL investigation does not take issue with qualitative analyses of small data samples. Conversely, the general trend nowadays is towards collecting big samples of data, which can then be analyzed statistically: if the sample is not big enough, the analysis will not be possible, and therefore the small amount of data negligible. While this is a perfectly acceptable viewpoint when large standardized languages are involved, it becomes problematic when investigating minority languages, languages with very few speakers. Paradoxically, GL is perfectly at ease with research on languages spoken by three speakers, in that it assumes that language is a psychological object, and therefore present in the mind of the speaker. Any insight into the products of human cognition can and should be investigated to gain a better understanding of how language works. As explicitly stated in D'Alessandro et al. (2021) and Leivada et al. (2019), less-spoken and endangered languages can reveal a great deal about possible and impossible structures, something that large standard languages may fail to show. Therefore, with respect to the importance attributed to less-spoken languages, GL is

perhaps a more equitable framework when compared with many centered in big-data collection, since the number of speakers is not a relevant factor for selecting structures to examine. We could summarize this point by saying that, exaggerating slightly, the *one speaker is enough*⁵ approach to linguistic study is an implicit recognition of the importance of minorities.

2.6 All languages are equal

If one speaker is enough (hyperbolically speaking), no large groups of speakers are needed to investigate a specific phenomenon, at least as it pertains to structural well-formedness. If two or three speakers producing a structure are enough to assume that a given construction is possible in a language, then the community of speakers does not necessarily need to be extremely large. This has immediate repercussions for the study of endangered and minority languages, which are often excluded by experimental investigations, since they do not reach the minimum number of controllable uniform stimuli that are necessary for statistics-based generalizations. In this sense, a methodological assumption can have positive implications on equality, as all speakers and all languages are equally informative and relevant for linguistic inquiry. It needs to be added that linguists working within the GL-framework are certainly not opposed to modeling variability using statistical methods. In fact, inter- and intra-speaker variabilities play a central role in current formal analyses, especially in research on L1 acquisition connected with statistical learning (Lidz and Gagliardi 2015; Yang 2016).

2.6.1 The ethical relevance of heritage languages

We stated that GL researchers never undermine the intuitions of any speaker, whether standard, non-standard/dialects, creoles or heritage languages, and instead use the data to answer questions bearing on the development of universal tendencies. This becomes particularly important in those cases in which languages are considered incomplete, or defective, like in the case of creoles (see Aboh 2016, Aboh and DeGraff 2022, McWhorter 2018, Mufwene 2001 and many others), and heritage languages (Montrul 2008; Polinsky 2005). As concerns heritage languages, they are very often considered incomplete, because of the fact that their speakers often

⁵ To be accurate, though, the *one speaker is enough* approach is, in practice, a relic of the past: the usual practice nowadays is to look for a small number of speakers sharing a grammar and relying on their intuitions. If a structure is produced systematically by only one speaker, it could be due to some idiolectal quirk. The habit according to which the researcher's intuitions suffice for the inquiry has also been abandoned.

receive qualitatively and quantitatively different amounts of input during the formative years of language acquisition. This view has been challenged by linguists who also work within generative frameworks, especially in light of the strong generative component within these studies, and ultimately dismissed on the basis of the “one-speaker-one language” approach discussed above (see also work by MacSwan 1999; Mahootian 1993). Recent studies in GL show indeed that heritage grammars are complex systems, and therefore warrant the same treatment as that given to the systems underlying monolingual/standard varieties (cf. D’Alessandro and Frasson 2023; D’Alessandro and Terenghi 2024; D’Alessandro et al. In pressb; Polinsky and Putnam 2024). We have seen in Section 2.4 that Hopp and Putnam (2015) demonstrate that rather than uniform paradigmatic leveling on the part of the remaining MSG-speakers to accept/produce sentences in which the finite verb occupies second position in all clauses, these remaining speakers have actually made their syntax “more complex” – especially with respect to the preferred placement of the finite verb in subordinate clauses. Similar conclusions have been reached by D’Alessandro and Frasson (2023), who have shown that Heritage Venetan speakers in Brazil have created an entirely new syntactic system for auxiliary selection, based on considerations like inverted subject position and anti-agreement effects. These auxiliaries, that in the Venetan spoken in Italy simply have a geographically determined distribution, have acquired in the heritage variety a specialization depending on highly complex syntactic configurations, showing once again that the grammar of HL speakers is not at all incomplete, less complex, or inferior in any way to full-fledged monolingual grammars. Once again, the study of minorities and highly fragmented languages has proven not only ethically sound but also scientifically relevant.

Of central importance here, GL considers heritage grammars as equally complete and worthy of investigation as any other grammatical system (D’Alessandro et al. 2021). This attitude has immediate repercussions on inclusion, as these languages are considered as important as any other, as well as on the status of these languages.

2.6.2 Microvariation

Regarding microvariation, the *Syntactic Atlas of Dutch Dialects* (SAND), the *Atlante Sintattico d’Italia* (ASIt),⁶ and the Microcontact Atlas⁷ are illustrative examples of projects that have successfully combined large-scale dialectal data collection with theoretical inquiries on variation. Research on minoritized and heritage languages

⁶ <http://asit.maldura.unipd.it/>.

⁷ <https://microcontact.hum.uu.nl/#home>.

has also gained pace in recent years (Benmamoun et al. 2013; Cummins 2005; Polinsky 2018). This theory-internal inquiry, along with the rise of cross-border migration, inter-mixing of populations and languages and the need for greater assimilation of communities, has opened up immense possibilities for GL to fruitfully combine empirical studies with theoretical research. This third methodological assumption of GL, *all languages are equal*, is often underestimated; however, we feel that this is one of the facets of generative inquiry that make it well suited to address ethical concerns in inquiry.

3 Drawbacks of the generative approach

Throughout the previous section, we highlighted the positive ethical aspects of adopting a GL methodology; however, some problems continue to persist in the generative methodology, also reflected in the literature. In most generative papers, usually no information is provided next to the examples other than the name of the language. The first obvious drawback of this, which appears much less in socially aware articles, is that the name linguists attribute to languages is not always the same that the linguistic community chooses for itself. There appears to be no immediate solution to this dilemma (see a recent debate between Haspelmath 2017; Dryer 2019 on who should decide the names of languages), which has been mentioned repeatedly with respect to the NLP methodology in D'Arcy and Bender 2023. Furthermore, labeling an example with the name of the language it supposedly represents contributes to other signs of inaccuracy, i.e., very often no other information other than the name is provided about the language, where it is spoken, by how many speakers, in which conditions. Providing this information requires little effort, but the additional detail improves the quality of the article tremendously, and could be easily implemented. We call this the *language labeling problem*, and we discuss this issue further in Section 3.1.

The second big drawback, which appears inevitable in studies that only target structures and sentences, is that the attitude of the speakers towards their language is almost completely disregarded. We already mentioned the importance that GL attributes to the speaker and their intuitions, and to the relevance of minority languages in understanding possible (and impossible) linguistic patterns. However, the excessive faith in the scientific and objective method may sometimes cause researchers to overlook the feeling the speakers have towards their language; should a speaker not want to associate themselves with a language, who decides that they are actually a speaker? We discuss speakers' attitudes in Section 3.2.

3.1 Language labeling

A linguistic methodology that puts a lot of attention on the structure and less on text, discourse, and the social use of language has no big need to discuss too much the social implications of using a label to refer to a language. Because of this, many generative articles place a label next to an example, and provide no further information regarding the language. Aside from a certain sloppiness, which might not have a direct impact on the data, we maintain that this methodology is also lacking from an ethical viewpoint. A disclaimer is necessary here: labeling is not an issue exclusive to GL. While this is obviously true, we find that GL articles are particularly prone to omitting basic information about languages and their users, and often use labels that are not even accepted by the speaker community. Nonetheless, this is a problem shared across linguistics proper. The first obvious problem is how to determine a language label. This is less of an issue for standardized varieties, but becomes very important for non-standardized ones. In what follows, we will present two examples of this problem: one regarding heritage Italo-Romance varieties in Latin America, and one regarding Hindi–Urdu spoken in India.

3.1.1 I am Italiano

The ERC project *Microcontact* investigated change in contact in heritage Italo-Romance varieties in the Americas (see Andriani et al. 2022a, 2022b; D'Alessandro et al. 2021, In pressa). The large inquiry targeted Italian Romance minority languages, both in Italy and in the Americas, namely in Argentina, Brazil, United States, and Canada. When faced with the decision of how to label examples and varieties, the research team decided to leave the first classification to the speaker. On the interactive atlas (<https://microcontact.hum.uu.nl/#home>), where speakers were supposed to upload some parts of their spoken language (with the help of the young in the community), the speakers were asked to classify their language variety by typing in its name. This classification overlapped almost always with the technical dialect classification of Romance languages in Italy, but in some instances that was not the case. We leave aside the issue of the attitude of the speaker, which we will return to in the next section, and concentrate on the language label. The question that the team members were faced with was if they had the right to decide to overwrite the language label and give it a different classification than that given by the speaker.⁸ It was decided to follow the standard classification and indeed overwrite the self-determined classification, but is that the right way to act?

⁸ For a similar problem, see D'Arcy and Bender (2023).

Taking the issue one step further, there was the case of speakers who call themselves Italian. While their nationality was undoubtedly so, the language they spoke was not Italian. Yet, many Italians claim to speak Italian, especially in large cities, like New York City, where many different dialects come into contact to form a koiné. This koiné was once called *broccolino* and is now simply believed to be Italian. Technically, it isn't. Or is it? Who decides? Another example of labeling/mislabeling comes from the language called Talian, which is a Venetan-based northern Italian variety spoken in Brazil. In Brazil, this language has gained official status (INDL 2014; Legislação Estadual do Rio Grande do Sul 2009), but in Italy, its base “lexifier”, Venetan, is not an official language. When interviewing Talian-Venetan speakers, the choice was between calling them Venetan or Talian. Most of the speakers identify themselves as Venetan, the official name of the language is however Talian. The choice we made was, in that case, to follow the speakers' intuitions and self-defined identity, and referred to the language as Heritage Venetan (see, for instance, D'Alessandro and Frasson 2023; Frasson et al. 2021). This entails, however, disregarding the official status of the language in Brazil, which was not what some of the speakers preferred. Since most speakers called themselves Venetan, we opted for this choice. However, if the researchers had to decide based on the speakers' choice, they would have had two sets of examples of the same language, even spoken within the same community, with two different names. In this case, following the speakers' self-identification would have created a fictitious division. While the solution to these kinds of dilemmas only depends on what the researcher gives prominence to, we consider it prudent not to take these issues for granted.

Another example of language labeling creating an ethical issue occurs in relation to Hindi–Urdu.

3.1.2 The Hindi–Urdu continuum

Hindi is the most prominent of India's official languages, and is also one of the most widely-studied Indo-Aryan languages in the generative literature. Generative linguists have extensively studied its grammatical features, including scrambling (Kidwai 2000; Mahajan 1990), clause structure (Dayal and Mahajan 2004), agreement (Bhatt 2005; Chandra 2007), etc. The alternative name ‘Hindi–Urdu’ is also used by linguists, since the language Urdu, typically associated with the Persian-influenced variety spoken by Muslims in north India, shares many grammatical properties with it, and also has a large vocabulary overlap. From a linguistic standpoint, therefore, the two languages are one and the same, though they are written in different scripts: Hindi uses the Devanagari script while Urdu uses the Nastaliq script.

The name-sharing is, however, confined to academic circles, with the government and the public still considering them as separate languages. In reality, the Hindi

and Urdu debate is a sensitive topic that often ignites communal tensions in the country, because of the alleged links between Hindi and Hindus, and between Urdu and Muslims. Despite being a widely used language in the Indian state of Uttar Pradesh (UP), it was only in 2014 that Urdu was finally given its constitutional rights as a second official language of UP, though, in practice, it is still to get its rightful place in education, media, and other official spaces (Chandra 2014, 2019). Urdu thus remains a largely marginalized language because of the lack of incentives available to its speakers. The promotion of Urdu is thus confined to certain economic groups and a small set of religious-educational institutions called *madrasas*.

Hindi–Urdu presents a very strange case, where linguists label Hindi and Urdu as one language, based only on their shared grammatical features. The complex geopolitics of the region, which to a certain extent forces the two to also retain separate identities, is largely overlooked by GL linguists. Because of this oversight, the grammatical differences between the two mostly remained unreported, resulting in lopsided research focusing on linguistic aspects of the more prominent and influential language, Hindi, a descendant of Sanskrit. There is very little theoretical research that reports data collected from the minority Muslim community and incorporates them as properties of Hindi–Urdu grammar. For instance, aside from the typical possessive constructions with the *kaa*-genitive (e.g., *Punjab-kaa sher* ‘the lion of Punjab’) reported for Hindi–Urdu, Urdu has an alternative *ezafe* construction (e.g., *sher-e-Punjab* ‘the lion of Punjab’); note the use of the *ezafe* marker *-e* as a linker and the reversed order of the possessor and the possessed nouns. There is some research on *ezafe* constructions by Bögel et al. (2009) and Chandra and Kumar (2013); however, in the mainstream generative literature, the *ezafe* form finds no mention in the grammar of Hindi–Urdu. This is one of many instances where grammatical nuances of Urdu go unnoticed in the description of Hindi–Urdu.

This problem can be addressed only with recognition of the different socio-cultural factors that prompt the languages to retain, develop, and sustain some unique grammatical characteristics. The *ezafe* form continues in Urdu due to the speakers’ exposure to Persian literature, an option that is mostly not available to Hindi speakers. If Hindi and Urdu are to be given a common Hindi–Urdu nomenclature, then features unique to Urdu should also be co-opted into descriptions and analyses of the Hindi–Urdu grammar. Generative linguists must develop a theoretical model that not only captures the similarities, but also explains how Persian-influenced features coexist alongside features that are typically associated with Sanskrit. To collect data that truly represents the Hindi–Urdu language, one must target all social groups claiming to speak either one or both of these languages (see Ahmad 2010 for fieldwork challenges for these languages). One should also take into consideration what the speakers of these languages think of a common name for the two.

3.2 The speakers' attitude

We already discussed the fact that sometimes the speakers have a different perception of the language they are speaking than the data collector. In most cases, this is of little impact, as the competing varieties have equal value in the eyes of the speakers. This is, however, not always the case. In this section we will report some illustrative examples of how a dilemma emerges between objectivity and accuracy and the speakers' attitude. Even more so than in the previous Section, 3.1, this is a general problem shared by all linguists.

3.2.1 Italian dialects or languages

Minority Italo-Romance languages are rather stigmatized both in Italy and in the Americas, where they are spoken as heritage languages. In Italy, the legacy of the fascist regime with respect to these minorities is still heavily ingrained in their speakers: Italo-Romance minority languages are considered inferior to Italian, they are still called *dialects* and they are to be avoided in formal contexts (with some regional exceptions). In general, speaking these languages is considered a sign of illiteracy rather than a welcome exhibition of bilingual knowledge (but see Garraffa et al. 2015, 2017), no education is provided in the languages (again, with some exceptions limited to Friuli Venezia Giulia and Sardinia), and they are slowly dying out.

A somewhat similar situation is found in the Americas, where the speakers of these languages, who left Italy before the spreading of standard Italian, carry with them an ambivalence resulting from speaking both a socially inferior language but yet the language linked to their identity. If fieldwork targets these minorities, speakers are often reticent and do not wish to claim that any of these is their main language. The options in this case are leaving the languages undocumented or just overriding the speakers' desire not to be associated to these languages. As we illustrate in the subsequent subsection, a similar situation is found in heritage speakers of Germanic languages in the US.

3.2.2 Heritage Germanic in the US

The documentation and study of heritage Germanic languages in the US has been the focus of linguistic research since the mid-19th century, exemplified by research on Danish (Grøngaard Jeppesen 2005), Dutch (Smits 1993; van Marle and Smits 1989; Webber 2011), German (Gilbert 1971), Icelandic (Arnbjörnsdóttir 2006), Norwegian

(Haugen 1969), and Swedish (Andreen 1900; Bergin 1909; Nelson 1943).⁹ Regarding these languages, Putnam (2024) makes an important distinction between two existing groups of Heritage Germanic speakers. He classifies one group as *Moribund Heritage Language speakers* (MHLs) and the other as *Diglossic Heritage Language speakers* (DHLs), where:

- *Moribund Heritage Language speakers (MHLs)*: Speakers of a heritage language of which the eldest or penultimate generation of speakers represent the final highly proficient speakers of the language. The heritage language is endangered, and in most cases moribund.
- *Diglossic Heritage Language speakers (DHLs)*: Speakers of a heritage language that continues to be transmitted as the L1 to children within the community. Most speakers are diglossic bi/multilinguals, possessing high proficiency in the sociolinguistically-dominant L2 of mainstream society, especially in particular domains.

The vast majority of research that has been carried out in Heritage Germanic languages-speaking communities has to date focused primarily on MHLs. The major proportion of DHLs are instead affiliated with conservative Anabaptist (Christian) groups, such as the Amish, Hutterites, and Old Order Mennonites. Although there is a healthy research tradition of language documentation and lexicography, many of these communities do not allow researchers to record them, based primarily on their religious beliefs related to the sin of recording or making a ‘graven image’ of them and their identity. The relationship that these groups have with modern technology is very fluid (Ems 2022; Kraybill et al. 2013), hence it is unwise to maintain rigid expectations in relation to ‘all’ Amish or ‘all’ Mennonites. Some will allow themselves to be recorded, and others will also participate in on-line experiments.¹⁰ Given that some members of these communities allow only partial data elicitation, one wonders whether assimilating them to the language is ethically valid.

Fieldwork techniques also need to be rethought when conducting research on non-standard and indigenous languages which go by different names and designations. Very often, we find cases where speakers themselves assume flexible linguistic identities, sometimes accommodating other languages with their original identity, and sometimes completely suppressing one linguistic identity with another. This

⁹ The interested reader is directed to the following sources for a more detailed overview of the history and current status of research on these varieties: Johannessen and Putnam (2020), Johannessen and Salmons (2021), and Putnam (2024). For our current purposes, we will home in on the challenges of working with these communities in connection with notions of ethical research.

¹⁰ Anecdotally, most conservative Anabaptists are more prone to participate in experiments if they can be ensured that their personal information (such as their names) are in no way associated with results.

makes the task of collecting native intuitions from speakers very difficult, since there is no clear-cut definition of what constitutes their native tongue. Generative linguists must be cognizant of the histories of these communities, so that they choose the right methods when collecting data from the speakers of these communities. Background sociological information is necessary here too, to aid us in choosing the correct target population and to improvise data collection methods.

3.2.3 The multiple identities of Rajbanshi

One curious case is that of an eastern Indo-Aryan language, Rajbanshi, which is variously identified as a separate indigenous language, a language related to Koch (previously discarded as belonging to a different tribe), and a variety of the standard Bangla language.

India has a notable number of indigenous languages that have not been accorded an official status by the Government of India. Many indigenous communities also remain socially vulnerable.¹¹ Rajbanshi is one such language spoken by an indigenous community in the state of West Bengal (and in adjoining Assam and Bangladesh).¹² It is an interesting instance of a language where its speakers have historically reinterpreted and rewritten their past to carve out new identities for themselves (cf. Debnath 2007; Nandi 2014).

Rajbanshi identity first became a serious issue in 1882, when the Indian Census Report categorized Rajbanshi and Koch as a single tribe. The Rajbanshis rejected this clubbing together, and instead claimed that Koch is an inferior (*Dalit*) caste that speaks an inferior language belonging to the Tibeto-Burman family, while the Rajbanshis belong to the higher (*Kshatriya*/warrior) caste, coming just behind the *Brahmins*. In 1926–1927, however, the community had a new narrative, when caste-based organizations submitted a requisition to the government to get the Rajbanshis recognized as *Dalits*, a demand that was finally granted to them in post-independent India. The shift was a necessary compromise for the community in order to gain

¹¹ From time to time, a tribal group may protest for better allocation of resources, more opportunities, and for an official recognition of its language, forcing the government to take notice and formulate a policy. This happened to the Austro-Asiatic language Santhali of the indigenous Santhal community that was co-opted into the Eighth Schedule of the Indian Constitution in 2003 after several efforts by the community members. Other languages, such as some aboriginal languages spoken in the Andaman and Nicobar islands, may be on the verge of extinction, mostly due to the lack of state support for the languages (Abbi 2020).

¹² In West Bengal, Rajbanshi, whose speakers are mainly found in the districts of Dinajpur, Malda, Jalpaiguri, and Cooch Behar, displays regional linguistic variation expected of any language spoken over such a large area. The language also goes by the names of Matribhasha, Kamtapuri, Surjapuri, and Koch.

access to government funds and resources reserved for the downtrodden castes and tribes. The Rajbanshis, who were once against the idea of being clubbed together with the Kochs, are now perfectly content with the new identity ‘Koch-Rajbanshi’ or ‘Kamata’. Additionally, some Rajbanshi speakers are standard Bangla bilinguals, either through access to higher education or merely due to contact with the latter. Rajbanshi identity is therefore a mix of multiple identities, where speakers identify themselves as Rajbanshi speakers, Rajbanshi-Koch speakers, or Rajbanshi-Bangla speakers.

Data collection with the purpose of writing a Rajbanshi grammar is a challenging task, as it is difficult to narrow in on an “idealized speaker” of Rajbanshi. The experience of one of the authors was that Rajbanshi exhibits an immense amount of variation, not just regionally, but also within local communities and families. Agreement markers, for instance, were noticeably different for different speakers: for some, the verbal endings sounded more Tibeto-Burman than Indo-Aryan. The fact that there are no historical records for the language makes the problem more serious, as no baseline grammar is available for comparison with the current language to help assess the changes that may have taken place due to contact with Koch and standard Bangla. One approach, adopted by the concerned author, was to use the knowledge of widely accepted/typical features of standard Bangla and Tibeto-Burman languages to filter out the seemingly borrowed cases, and use the rest and most commonly attested patterns to characterize the original Rajbanshi. While this method is not foolproof, it serves as a first step towards identifying the essential features of the language not yet influenced by the other languages. It is worth emphasizing again, that in order to decide which method works best for data collection and analysis of Rajbanshi, linguists must first be aware of its history and other socio-cultural factors that led to the creation of fluid linguistic identities for its speakers. Additionally, the perception of the speaker regarding her language and linguistic identity should be noted down carefully so that any idiolectal difference can be explained in the right context, with reference to the other languages (standard Bangla, Koch) that the speaker may conflate with Rajbanshi.

4 Conclusions

In this paper, we have presented some existing good practices of GL research and also identified factors that, if GL practitioners could become more aware of them, would further strengthen their theoretical analyses. Many of the issues identified are not only relevant for GL, but can also be recognized in many others, if not all, approaches to linguistic research. The core methodological assumptions of GL that include concepts such as idealized speaker–hearer, native intuitions of a speaker,

one speaker is enough, and all languages are equal have proven to be very useful in the formulation of universal, combinatorial tools for language. They also provide space for minority and non-standard languages, some of which have very few speakers. There is, however, a lot of scope for improvement and for making the GL enterprise more inclusive and appreciative of understudied languages and their speakers. Languages vary depending on a number of factors including regional and socio-cultural differences. Additionally, speakers sometimes have different perceptions of their languages and linguistic identities than the ones linguists hold, they may be more comfortable with one linguistic identity or with multiple ones, all speakers may not have the same level of proficiency in their language etc.; all these grammar-external factors contribute to the nature of data collected from the speakers, their theoretical analyses, and the manner in which linguists classify, name, and present these languages to the academic world. Keeping these issues in mind, we believe that it will help GL research if theoretical linguists become more mindful of the speakers, their linguistic proficiency, their attitudes towards their languages, and overall the histories and societal factors that shape their opinions.

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